

Trabelsi Slim

Mechanical Engineering Student | Design, Simulation & System Reliability

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🌐 Open to relocation 📅 Applying for a Final-Year Internship (PFE) ✅ Available from February 2027

🇫🇷 Anglais (TOEIC B2) | 🇫🇷 Français (DELFI B2)

PROFESSIONAL SUMMARY

Designs and improves the reliability of technical systems with a focus on performance: achieved a **15% improvement in the optical efficiency** of a solar thermal concentrator, validated through **20+ simulations**, and delivered **10+ custom fabrication** projects with zero rework.

SKILLS

— **CAD & Simulation:** SolidWorks, CATIA, Abaqus, Ansys Mechanical, MATLAB

— **Programming & Tools:** Python, C, C++, Arduino, ESP32, JavaScript, HTML, CSS

EXPERIENCE

Manufacturing Technician & CAD Modeler (Part-time)

06/2020 – Present | Tunisia

Future Pub – Metal Structures and Manufacturing

- Delivered **10+ custom fabrication** projects alongside studies, with **100% on-time** delivery and zero rework; modeled parts in **SolidWorks**.
- Operated **MAG welding**, lathe, and milling equipment to manufacture components to **±0.5 mm** tolerances, and managed on-site assembly for **10+ client job sites**.

Engineering Intern – Compressed Air Systems

07/2026 – 08/2026 | Tunisia

Marquardt Tunisia [🔗](#)

- Analyzed the plant's compressed air network, identifying **3+ critical efficiency gaps** and writing structured **technical reports**.
- Developed **2 action plans** to reduce energy fluctuations; standardized diagnostics to speed up data collection.

Engineering Intern – Industrial Maintenance

07/2025 | Tunisia

Grande Fabrique De Confiserie Orientale (GFCO)

- Oversaw **15+ preventive and corrective maintenance interventions** across 3 active production lines, reducing unplanned downtime through root-cause analysis.
- Authored a **maintenance procedures document** adopted by the team to optimize production scheduling and reduce intervention lead times.

PROJECTS

Year-End Project – Solar Thermal Concentrator

2025 – 2026

National Engineering School of Tunis (ENIT)

- Achieved a **15% improvement** in optical efficiency through geometric optimization of a compound parabolic concentrator, validated by **20+ ray-tracing simulations**; built a complete **CAD model in SolidWorks** that reduced prototyping time.
- Conducted a techno-economic analysis projecting a heat cost of 0.091–0.098 TND/kWh and a **14-year payback period**; developed **Python** and **JavaScript** tools that reduced analysis time.

Finite Element Analysis – Structural Study

2025 – 2026

Academic Project (ENIT)

- Modeled **3+ mechanical components** in **CATIA** and ran structural simulations in **Abaqus** and **Ansys Mechanical** to evaluate stress, strain, and displacement under service conditions.
- Produced **3 technical reports** with verified failure analysis, material recommendations, and geometric modifications.

VOLUNTEERING & INVOLVEMENT

Board Member & Technical Lead

2024 – 2026 | Tunis, Tunisia

Robotics Club (ENIT)

- Led a national robotics competition with **100+ participants**, overseeing logistics, recruitment, and execution, and mentored **junior engineering students** through **5+ technical workshops**.
- Designed and programmed **2 autonomous robots** from concept to competition (all-Terrain and line-follower); competed in **2 national competitions**.

FORMATION

Engineering Degree – Mechanical Engineering

09/2024 – Present | Tunis, Tunisia

National Engineering School of Tunis (ENIT) [🔗](#)